CrystalSight: language models read a crystal's text well, but not its 3D space

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> **Key finding.** When a question about a crystal needs real 3D work (enumerating symmetry copies, following a hydrogen-bond network, counting neighbours), text-only reasoning collapses. From CIF text, two current models score 0.20–0.43, and from rendered images or an interactive ASCII 3D viewer 0.10–0.22 (random: 0.13). Give the same models small geometry tools and both reach about 0.75.

Question. LLM agents are asked to check crystal structures, shown to them as CIF text, pictures, an interactive viewer or analysis tools. Does that choice decide what they can work out?

Two sets of spatial questions (20 each, five families of four). The *easy set* (v3) needs one contact or one plane: a donor–acceptor symmetry relation, the H-bond motif, nearest neighbours, a ring's orientation, π-stacking. The *hard set* (v4) needs exact enumeration or graph search, with atoms named in a copy-aware form `SITE#k[i,j,k]` (symmetry operation k, lattice shift [i,j,k]). Its five families are:

- **G1:** the acceptor copy nearest a given hydrogen;
- **G2:** the number of H-bond steps from a molecule to its own lattice translate;
- **G3:** the size of the smallest H-bond ring through a given bond, and how many molecules it spans;
- **G4:** how many molecules lie within a radius;
- **G5:** neighbour counts in the ordered major and minor replicas of a disordered structure.

Structures come from the Crystallography Open Database and MolCrysKit examples, plus a few licensed CSD entries. The CSD entries are used locally under anonymous ids and are not redistributed.

Gold answers from geometry. Gold answers are computed from symmetry-expanded coordinates. A hydrogen bond is D–H···A with D and A in {N, O}, H···A ≤ 2.5 Å and an angle of at least 120°. Every answer is recomputed with all cutoffs shifted by ±5 %, and questions whose answer changed were replaced. Partial credit per JSON field; a copy label earns credit for site, operation and shift separately.

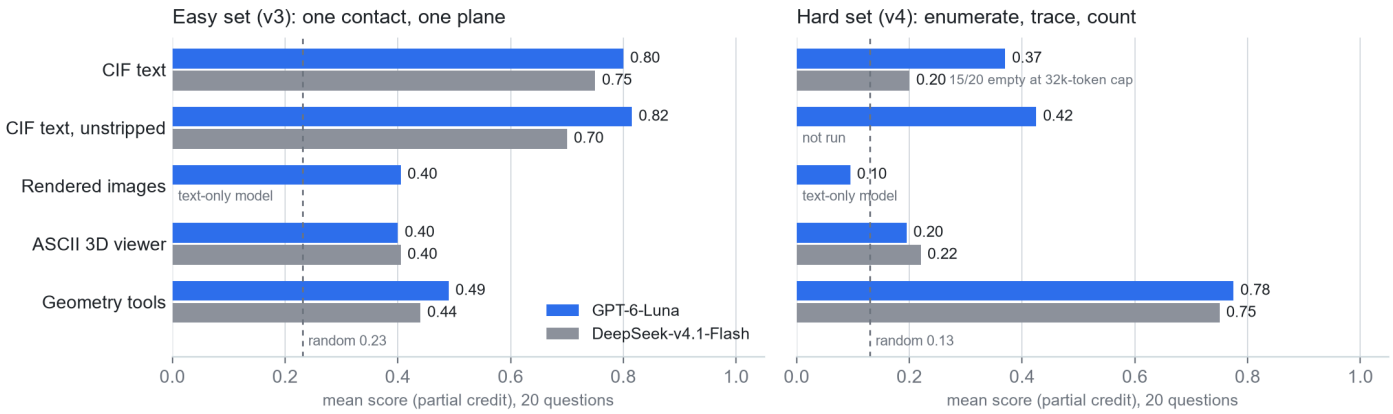
Five channels, one at a time:

- CIF text with publication metadata stripped;
- unstripped CIF, as a leakage control;
- three rendered images;
- an interactive MatterVis ASCII 3D viewer (rotate, zoom, relabel, with an H-bond overlay, distance query and disorder-component switch);
- **geometry tools**: MolCrysKit plus six small tools built for this benchmark (H-bond list, nearest atoms, H-bond path, H-bond ring, neighbour shell, disorder components). The tools return raw lists, which the model must still count and interpret.

The models are GPT-6-Luna and DeepSeek-v4.1-Flash (text only), at temperature 0 with up to 32k output tokens.

On hard spatial questions, text reasoning collapses and only geometry tools hold up

Same structures and questions, five ways of showing them. Easy set: answers need one contact or one plane. Hard set: answers need enumeration, H-bond graph search or exact counts.



Results. On the easy set, CIF text alone scores 0.70–0.82, and images and the viewer about 0.40. On the hard set, GPT-6-Luna's text scores fall to 0.37 (CIF) and 0.43 (unstripped CIF). DeepSeek returned no answer to 15 of 20 CIF questions: it reasoned until the 32k-token limit. Images (0.10) and the viewer (0.20–0.22) stay near chance. With geometry tools, both models reach 0.75–0.78 (strict 0.75).

The heatmap shows where text survives. G1, a single nearest contact, is solved from text (0.85–1.00). Paths, rings and counts mostly are not, and disorder replicas defeat every channel, tools included (0.25–0.38).

Without tools, only the single-contact question (G1) is reliably solved

Hard set (v4), mean score per question family (4 questions each). DeepSeek CIF text: old text input, 15/20 replies empty at the token cap.

	Nearest partner (copy label)	H-bond hops to a translate	Smallest H-bond ring	Neighbour-shell count	Disorder replicas
Luna · CIF text	0.85	0.50	0.25	0.25	0.00
Luna · CIF text, unstripped	1.00	0.25	0.62	0.25	0.00
Luna · Rendered images	0.10	0.00	0.00	0.25	0.12
Luna · ASCII 3D viewer	0.60	0.25	0.00	0.00	0.12
Luna · Geometry tools	1.00	0.75	1.00	0.75	0.38
DeepSeek · CIF text	1.00	0.00	0.00	0.00	0.00
DeepSeek · ASCII 3D viewer	0.60	0.25	0.00	0.25	0.00
DeepSeek · Geometry tools	1.00	1.00	1.00	0.50	0.25

Leak found and removed. Many published CIFs include a precomputed hydrogen-bond table with symmetry codes. That table gives away G1 answers and the edges of the G2/G3 network, so for the hard set we remove it from both text channels. We also fixed a text-stripping bug that dropped the `loop_` keyword from CIF tables. GPT-6-Luna's text channels were re-run on the clean input (CIF 0.31 → 0.37, unstripped CIF 0.43 → 0.43). DeepSeek's CIF result and all easy-set text results still use the old input.

Limitations. 20 questions per set, 2 models, one run each: one question moves a score by 0.05. The tools were written for these question types, so they show what the right tools make possible, not a general agent. An earlier 247-question set of structure checks (missing H, disorder, coordination) showed the same ordering: text above images and the viewer. Code, tasks, gold and scored results are in the repository.